testbench.sv

```
module full_adder (
    input A, B, Cin,
    output Sum, Cout
);
    assign Sum = A ^ B ^ Cin;


    assign Cout = (A & B) | (Cin & (A ^ B));
endmodule

module adder_4bit_rca (
    input [3:0] A,
    input [3:0] B,
    input Cin,
    output [3:0] Sum,
    output Cout
);
    wire c1, c2, c3;

    full_adder fa0 (.A(A[0]), .B(B[0]), .Cin(Cin), .Sum(Sum[0]), .Cout(c1));
    full_adder fa1 (.A(A[1]), .B(B[1]), .Cin(c1),  .Sum(Sum[1]), .Cout(c2));
    full_adder fa2 (.A(A[2]), .B(B[2]), .Cin(c2),  .Sum(Sum[2]), .Cout(c3));
    full_adder fa3 (.A(A[3]), .B(B[3]), .Cin(c3),  .Sum(Sum[3]), .Cout(Cout));

endmodule
```

design.sv

```
`timescale 1ns / 1ps

module tb_adder_4bit;
    reg [3:0] A;
    reg [3:0] B;
    reg Cin;

    wire [3:0] Sum;
    wire Cout;

    adder_4bit_rca uut (
        .A(A),
        .B(B),
        .Cin(Cin),
        .Sum(Sum),
        .Cout(Cout)
    );

    initial begin
        $monitor("Zaman=%0t | A=%b, B=%b, Cin=%b | Sum=%b, Cout=%b", $time, A, B, Cin, Sum, Cout);

        A = 4'b0000; B = 4'b0000; Cin = 0; #10;

        A = 4'b0101; B = 4'b0011; Cin = 0; #10;

        A = 4'b1111; B = 4'b0001; Cin = 0; #10;

        A = 4'b1010; B = 4'b0101; Cin = 1; #10;

        A = 4'b1111; B = 4'b1111; Cin = 1; #10;

        $finish;
    end
endmodule
```

```
design.sv:1: ...: The inherited timescale is here.
Zaman=0 | A=0000, B=0000, Cin=0 | Sum=0000, Cout=0
Zaman=10000 | A=0101, B=0011, Cin=0 | Sum=1000, Cout=0
Zaman=20000 | A=1111, B=0001, Cin=0 | Sum=0000, Cout=1
Zaman=30000 | A=1010, B=0101, Cin=1 | Sum=0000, Cout=1
Zaman=40000 | A=1111, B=1111, Cin=1 | Sum=1111, Cout=1
design.sv:32: $finish called at 50000 (1ps)
```